Chapter 3. Connectedness and Compactness

Connected and path-connected spaces, components, compact and limit point compact spaces, local compactness

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3.1 Connected spaces

Definition (Connectedness)

A **separation** of X is a pair of disjoint nonempty open sets with union X. X is **connected** if it has no separation, equivalently if the only sets that are both open and closed are $\emptyset$ and X.

Lemma 3.1

If C, D separate X and $Y \subset X$ is connected, then $Y \subset C$ or $Y \subset D$.

Theorem 3.2

(a) A union of connected sets with a common point is connected. (b) If A is connected and $A \subset B \subset \overline{A}$, then B is connected. (c) Continuous images of connected spaces are connected. (d) Finite products of connected spaces are connected; so is any product in the product topology.

Proof. (a), (b) follow from the lemma. (c) A separation of $f(X)$ pulls back. (d) For $X \times Y$: fix $b \in Y$; for each x, $T_x = (X \times \{b\}) \cup (\{x\} \times Y)$ is connected (two connected sets sharing (x, b)), all T_x contain $X \times \{b\}$, and their union is $X \times Y$; apply (a). For arbitrary products, the points differing from a fixed point a in finitely many coordinates form a connected dense set. ■

$\mathbb{R}^\omega$ with the box topology is not connected: bounded and unbounded sequences separate it.

3.2 Connected subspaces of the real line

A simply ordered set L with more than one element is a **linear continuum** if it has the least upper bound property and between any two elements there is a third.

Theorem 3.3

A linear continuum, and every interval or ray in it, is connected. In particular $\mathbb{R}$ and its intervals are connected.

Proof. Suppose Y convex and A, B separate Y, with $a \in A$, $b \in B$, $a < b$. Let $c = \sup(A \cap [a, b])$. If $c \in B$, then (as B is open in Y) some interval $(d, c] \subset B$, giving a smaller upper bound d. If $c \in A$, then $c < b$ and some $[c, e) \subset A$, so c is not an upper bound. Either way a contradiction. ■

Theorem 3.4 (Intermediate value theorem)

If $f : X \to Y$ is continuous, X connected, Y ordered, and $f(a) < r < f(b)$, then $r = f(c)$ for some c.

X is **path connected** if any two points are joined by a path $[0, 1] \to X$. Path connected spaces are connected; the converse fails:

Example 3.1

(a) The **topologist's sine curve** $\overline{S}$, the closure of $\{(x, \sin(1/x)) : 0 < x \leq 1\}$, is connected (closure of a connected set) but not path connected: a path from $(0, 0)$ into S would, at the first time it leaves $\{0\} \times [-1, 1]$, be forced to oscillate.

(b) The **long line** $L = S_\Omega \times [0,1)$ (dictionary order) is a path connected linear continuum which is locally homeomorphic to $\mathbb{R}$ but not imbeddable in $\mathbb{R}^n$.
(c) $\mathbb{R}^n - \{0\}$ $(n \geq 2)$ and S^{n-1} are path connected; $\mathbb{R}$ and $\mathbb{R}^2$ are not homeomorphic (remove a point).

3.3 Components and local connectedness

The **components** of X are the equivalence classes of "lying in a common connected subspace"; they are connected, disjoint, closed, and every connected subspace lies in one. **Path components** are defined with paths. X is **locally (path) connected** if every neighbourhood of every point contains a (path) connected neighbourhood.

Theorem 3.5

X is locally connected iff components of open sets are open. If X is locally path connected, its components and path components coincide.

Proof. For the second statement: path components of a locally path connected space are open; the path component P of x is then also closed (its complement is a union of path components), so it equals the component of x. ■

The topologist's sine curve is connected but not locally connected; $\mathbb{Q}$ has singleton components and is not locally connected.

3.4 Compact spaces

X is **compact** if every open cover has a finite subcover.

Theorem 3.6

(a) Closed subspaces of compact spaces are compact. (b) Compact subspaces of Hausdorff spaces are closed; in fact a compact set and a point outside it have disjoint neighbourhoods. (c) Continuous images of compact spaces are compact. (d) A continuous bijection from a compact space to a Hausdorff space is a homeomorphism.

Proof. (b) For $x_0 \notin Y$, choose disjoint $U_y \ni y$, $V_y \ni x_0$ for $y \in Y$; finitely many U_y cover Y, and the intersection of the corresponding V_y is a neighbourhood of x_0 missing their union. (d) Closed sets map to compact, hence closed, sets. ■

Lemma 3.7 (Tube lemma)

If Y is compact and N is an open set in $X \times Y$ containing $\{x_0\} \times Y$, then N contains a tube $W \times Y$ with W a neighbourhood of x_0.

Proof. Cover $\{x_0\} \times Y$ by finitely many basis sets $U_i \times V_i \subset N$ and take $W = \bigcap U_i$. ■

Hence finite products of compact spaces are compact (Tychonoff's theorem for infinite products is Chapter 5). X is compact iff every collection of closed sets with the **finite intersection property** has nonempty intersection.

3.5 Compact subspaces of the real line

Theorem 3.8

In an ordered set with the least upper bound property, closed intervals $[a, b]$ are compact. A subspace of $\mathbb{R}^n$ is compact iff it is closed and bounded (Heine–Borel).

Proof. Given an open cover of $[a, b]$, let C be the set of $y > a$ such that $[a, y]$ is finitely covered. C is nonempty and $c = \sup C$ belongs to some member U of the cover, which contains an interval $(d, c]$; so $c \in C$, and if $c < b$, U also covers points beyond c. Hence $b \in C$. Then $\mathbb{R}^n$: a closed bounded set lies in a product $[-M, M]^n$. ■

Theorem 3.9 (Extreme value theorem)

A continuous $f : X \to Y$, X compact and Y ordered, attains a maximum and a minimum.

Lemma 3.10 (Lebesgue number lemma)

If $\mathcal{A}$ is an open cover of a compact metric space X, there is $\delta > 0$ such that every set of diameter less than δ lies in some member of $\mathcal{A}$.

Proof. Choose a finite subcover $A_1, \ldots, A_n$ (all $\neq X$), let $C_i = X - A_i$ and $f(x) = \frac{1}{n} \sum d(x, C_i)$; $f > 0$ and continuous, so $f \geq \delta > 0$. If x has $d(x, C_m) = \max_i d(x, C_i) \geq f(x) \geq \delta$, the ball $B(x, \delta) \subset A_m$. ■

Consequences: continuous maps from compact metric spaces are uniformly continuous; a nonempty compact Hausdorff space without isolated points is uncountable.

3.6 Limit point compactness and local compactness

X is **limit point compact** if every infinite subset has a limit point, and **sequentially compact** if every sequence has a convergent subsequence.

Theorem 3.11

Compact implies limit point compact. For metrizable spaces, compact, limit point compact and sequentially compact are equivalent.

Proof. If A has no limit point, A is closed and each point has a neighbourhood meeting A in one point; these neighbourhoods with $X - A$ cover X, so A is finite. For metric X, sequential compactness gives a Lebesgue number and total boundedness (X is covered by finitely many ϵ-balls), and these together give compactness. ■

In general these notions differ: S_Ω is limit point compact and sequentially compact but not compact.

X is **locally compact** at x if some compact set contains a neighbourhood of x. $\mathbb{R}^n$ is locally compact; $\mathbb{Q}$ and $\mathbb{R}^\omega$ are not.

Theorem 3.12 (One-point compactification)

A space X is locally compact Hausdorff iff there is a compact Hausdorff Y with $X \subset Y$ and $Y - X$ a single point; Y is unique up to homeomorphism. (Open sets of Y: open sets of X, and $Y - C$ for compact $C \subset X$.)

For example $\mathbb{R}^n \cup \{\infty\} \cong S^n$ by stereographic projection. In a Hausdorff space, local compactness at x is equivalent to: every neighbourhood U of x contains a neighbourhood V with $\overline{V}$ compact and $\overline{V} \subset U$.